

Subgroup Performance

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Improve Generalization

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Learning Curves

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Metrics Match Consequences

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Check sensitivity for Subgroup evaluation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Subgroup evaluation. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the subgroup gap. What causal claim is not supported?

6. Solve the given case

A has 18/20 correct; B has 6/10.

Compute subgroup results and evidence limits.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Accuracy is 18/20 in bright light and 6/10 in dim light. Compute both rates and the percentage-point gap.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 31 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Add subgroup tables and a minimum acceptable threshold to the Plant Classifier release criteria.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the subgroup gap. What causal claim is not supported?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Calculate group rates

Calculate the worked group and pooled rates.

My reasoning and evidence

13. W2 Check sensitivity

How does correcting one B error affect its rate?

My reasoning and evidence

14. W3 Report missing coverage

Group C has no evaluated cases. What is its measured error rate?

My reasoning and evidence

15. W4 Apply independently

Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

My reasoning and evidence

16. W5 Avoid a fairness shortcut

Why does equal accuracy not settle whether use is fair?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Subgroup Performance

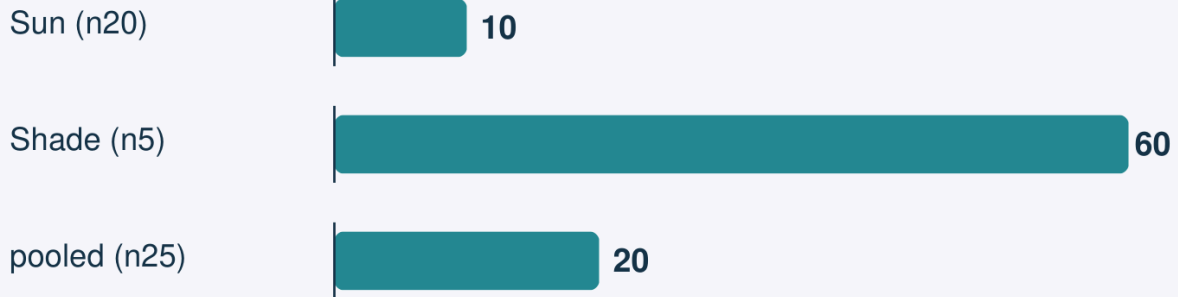
Calculate subgroup and pooled error rates without letting the larger group hide a gap.

Group Sun: 2 errors in 20 photos; Shade: 3 errors in 5. Same error definition; related plant views are documented.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Subgroup error rates

percent



Bar length shows the supplied value; the marked baseline is zero.

Subgroup Performance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Sun rate

Sun error= $2/20=10\%$.



2. Shade rate

Shade error= $3/5=60\%$.



3. Pooled rate

Total error= $5/25=20\%$, which hides the 50-percentage-point subgroup difference.

CHANGE ONE THING

Shade gains 5 new cases with no errors.

Explain the changed result using the matching lesson.

Why should the report retain subgroup rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Calculate subgroup and pooled error rates without letting the larger group hide a gap.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Total error= $5/25=20\%$, which hides the 50-percentage-point subgroup difference.

1. Pooled rate
2. Sun rate
3. Shade rate

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Pooled rate Total error= $5/25=20\%$, which hides the 50-percentage-point subgroup difference. Read the picture in order. Total error= $5/25=20\%$, which hides the 50-percentage-point subgroup difference. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Shade gains 5 new cases with no errors. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Shade error becomes $3/10=30\%$; pooled error becomes $5/30\approx 16.7\%$.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Shade error becomes $3/10=30\%$; pooled error becomes $5/30\approx 16.7\%$. Extra evaluated cases change denominators; existing errors have not disappeared.

Repair a missing building block

Review Improve Generalization (ai-4/30). Goal: Compare training and validation evidence for model capacity. Short practice: What fresh evidence is needed after adding crossed training data? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Learning Curves (ai-4/28). Goal: Read training-size learning curves. Short practice: What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

